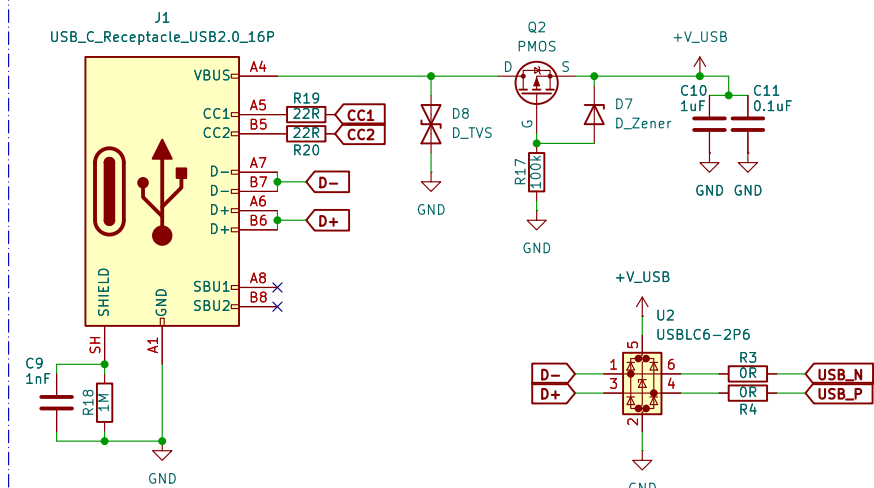


USB-C POWER SUPPLY 5 / 9 / 12 / 20 / 28V

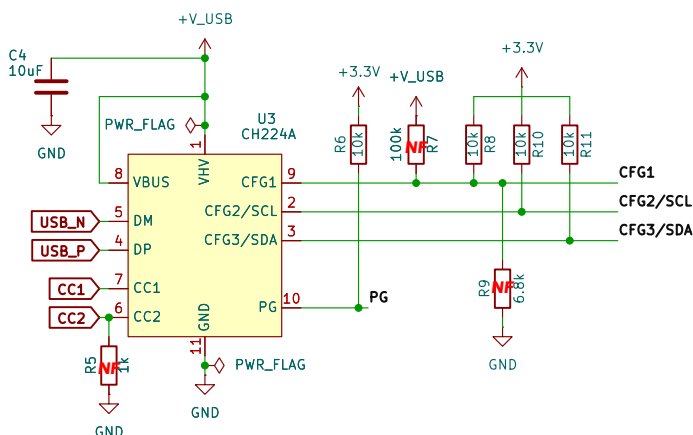
USB-C INPUT



DESIGN NOTE: CH224A already has the USB sink detection circuitry internally. Thus, external 5.1kR resistors are not needed on the CC1 and CC2 lines.

To request output greater than 20V or greater than 60W, use a Type-C male connector and connect a 1 kΩ resistor from CC2 to GND. This is a special mode called eMarker simulation.

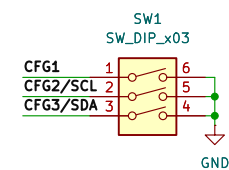
USB PD CONTROLLER



DESIGN NOTE: The unpopulated resistors on CFG1 are for experimenting single resistor configuration and ext. pull-up.

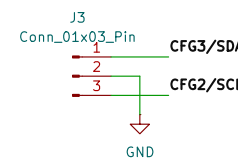
DESIGN NOTE: Pin PG is Active-Low by default. Thus, externally pulled-up.

OUTPUT V SELECTION SWITCHES

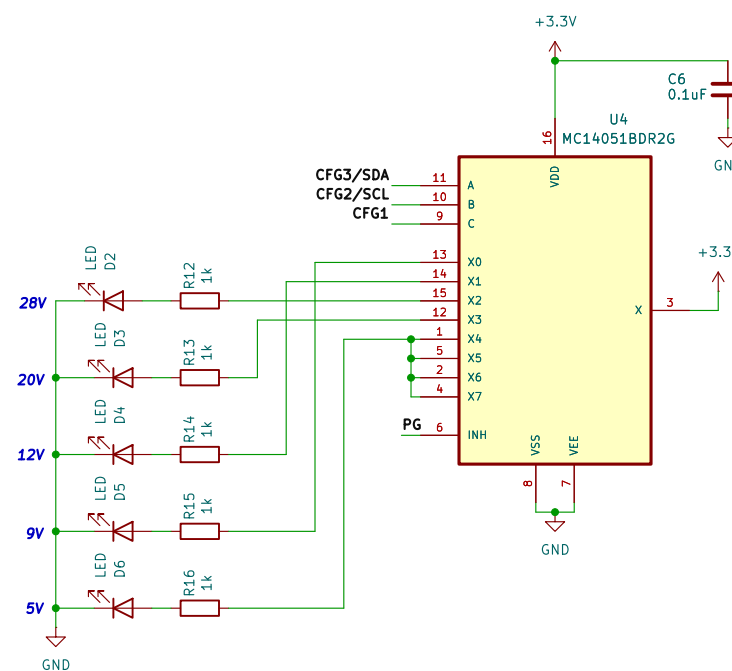


CFG1	CFG2	CFG3	Output V
1	X	X	5V
0	0	0	9V
0	0	1	12V
0	1	1	20V
0	1	0	28V

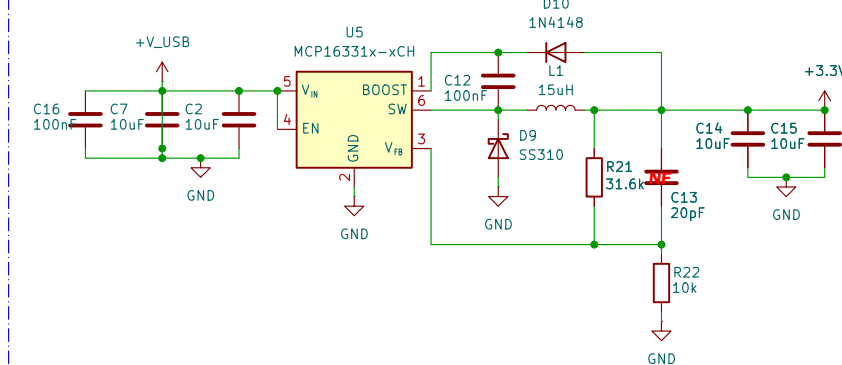
I2C CONNECTOR



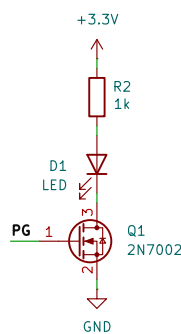
OUTPUT VOLTAGE INDICATOR LEDS



+3.3V POWER

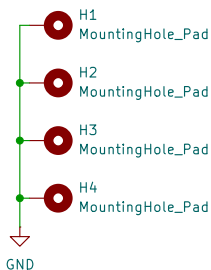


ERROR INDICATOR

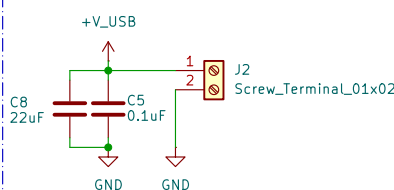


DESIGN NOTE: Proper working gives PG=0 (Active-Low). If there is an error PG wouldn't be 0 and turns this NOT MOSFET and the LED lights up.

MOUNTING HOLES



OUTPUT CONNECTOR



IMPORTANT! The maximum output current and voltage depend on the power adapter, cable and the selected voltage.

Sheet: /
File: USB-C_Power_Supply.kicad_sch

Title:

Size: A3 Date: 2026-06-11

Rev: V1.0
Id: 1/1